Some Details

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1 Normalization of Polarization Vectors

First, we give the definition of polarization vectors which is written in terms of adjoint representation

$$\varepsilon_a^\mu = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p^J\rangle(\sigma_a)^I{}_J}{4(p\cdot q)} \tag{1}$$

here a runs from 1 to 3. We can use antisymmetric tensor to rewrite this equation,like

$$\varepsilon_a^\mu = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p_K\rangle\epsilon^{JK}(\sigma_a)^I{}_J}{4(p\cdot q)} \tag{2}$$

here we choose the left-multiplication convention.

Then, a 2×2 symmetric matrix can be defined like

$$(\sigma_a)^{IJ} = \epsilon^{JK}(\sigma_a)^I{}_K. \tag{3}$$

This means that the original polarization vector can be understood as

$$\begin{aligned}\varepsilon_a^\mu &= -\frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p_J\rangle}{4(p\cdot q)}(\sigma_a)^{IJ} \\ &= -\varepsilon_{IJ}^\mu(\sigma_a)^{IJ}\end{aligned} \tag{4}$$

here I,J are fundamental little group indices, runs from 1 to 2.

And we know that the reasonable polarization vector for physical state should be symmetric with little group indices, it means that we need to symmetrize the indices I and J as $(IJ) = \frac{1}{2}(IJ+JI)$. Because of the symmetry of $(\sigma_a)^{IJ}$, so it can be obtained that

$$\varepsilon_a^\mu = -(\varepsilon_{(IJ)}^\mu + \varepsilon_{[IJ]}^\mu)(\sigma_a)^{IJ} = -\varepsilon_{(IJ)}^\mu(\sigma_a)^{IJ} \tag{5}$$

which means we only need to consider the symmetric part of ε_{IJ}^μ.

Now, we want to compute the normalization for ε_{IJ}^μ, and after multiplying the symmetric matrix, it should go back to the normalization

$$\varepsilon_a^\mu\varepsilon_{b,\mu} = -\delta_{ab} \tag{6}$$

The method to compute the normalization is just multiplying two polarization vectors directly, like

$$\varepsilon^{\mu}_{(IJ)}\varepsilon_{(KL),\mu}=\frac{\langle p_I|(r\cdot\Gamma)\Gamma^{\mu}|p_J\rangle}{4(p\cdot q)}\times\frac{\langle p_K|(r\cdot\Gamma)\Gamma_{\mu}|p_L\rangle}{4(p\cdot q)} \tag{7}$$

here we need to use a special 5d equality

$$(\Gamma^{\mu})_A^{\ B}(\Gamma_{\mu})_C^{\ D}=-2\Omega_{AC}\Omega^{BD}+2\delta_A^{\ D}\delta_C^{\ B}-\delta_A^{\ B}\delta_C^{\ D} \tag{8}$$

and the dirac operator can be decomposed to

$$\not{p}=p\cdot\Gamma=p_A^{\ B}=|p_I\rangle_A\langle p^I|^B \tag{9}$$

which help us to rewrite the equation (7) to be

$$\begin{aligned}\varepsilon^{\mu}_{IJ}\varepsilon_{KL,\mu}&=\frac{\langle p_I|^A|r_M\rangle_A\langle r^M|^B(\Gamma^{\mu})_B^{\ C}|p_J\rangle_C\langle p_K|^D|r_N\rangle_D\langle r^N|^E(\Gamma_{\mu})_D^{\ F}|p_L\rangle_F}{16(p\cdot q)^2}\\&=\frac{\langle p_I|r_M\rangle\langle p_K|r_N\rangle}{16(p\cdot q)^2}\times\langle r^M|^B(\Gamma^{\mu})_B^{\ C}|p_J\rangle_C\langle r^N|^E(\Gamma_{\mu})_E^{\ F}|p_L\rangle_F\end{aligned} \tag{10}$$

After plugging the equation (8) into the second part of (10), it becomes

$$\begin{aligned}&\langle r^M|^B|p_J\rangle_C\langle r^N|^E|p_L\rangle_F(-2\Omega_{BE}\Omega^{CF}+2\delta_B^{\ F}\delta_E^{\ C}-\delta_B^{\ C}\delta_E^{\ F})\\&=-2\langle r^M|r^N\rangle\langle p_L|p_J\rangle+2\langle r^M|p_L\rangle\langle r^N|p_J\rangle-\langle r^M|p_J\rangle\langle r^N|p_L\rangle\end{aligned} \tag{11}$$

The first term of (11) equals to 0 because the equality

$$\langle p_I|p_J\rangle=0 \tag{12}$$

and the remaining two term can be contracted with the first part of (10), which helps to obtain the following two terms

$$2\langle p_I|r_M\rangle\langle r^M|p_L\rangle\langle p_K|r_N\rangle\langle r^N|p_J\rangle-\langle p_I|r_M\rangle\langle r^M|p_J\rangle\langle p_K|r_N\rangle\langle r^N|p_L\rangle \tag{13}$$

It is known that

$$\langle p_I|(r\cdot\Gamma)|p_J\rangle=\mathrm{Tr}[|p_J\rangle\langle p_I|(r\cdot\Gamma)]=\frac{1}{2}\epsilon_{IJ}\mathrm{Tr}[(p\cdot\Gamma)(r\cdot\Gamma)]=2\epsilon_{IJ}(p\cdot r) \tag{14}$$

so the formal equation can be rewritten like

$$8(p\cdot r)^2\epsilon_{IL}\epsilon_{KJ}-4(p\cdot r)^2\epsilon_{IJ}\epsilon_{KL} \tag{15}$$

Finally, we need to symmetrize the indices (IJ) and (KL), so that we obtain

$$\varepsilon^{\mu}_{IJ}\varepsilon_{KL,\mu}=\frac{1}{8(p\cdot r)^2}(p\cdot r)^2(\epsilon_{IL}\epsilon_{KJ}+\epsilon_{JL}\epsilon_{KI}+\epsilon_{IK}\epsilon_{LJ}+\epsilon_{JK}\epsilon_{LI})-\frac{1}{16(p\cdot r)^2}(p\cdot r)^2(\epsilon_{(IJ)}\epsilon_{(KL)}) \tag{16}$$

because ϵ itself is an antisymmetric tensor so it will become zero under the symmetrization, so that the second term is vanishing.

After some rearrangement of indices, we can obtain

$$\varepsilon^{\mu}_{IJ}\varepsilon_{KL,\mu}=\frac{1}{4}(\epsilon_{IL}\epsilon_{KJ}+\epsilon_{IK}\epsilon_{LJ}) \tag{17}$$

2 Completeness Relation of Polarization Vectors

For this purpose, we need to compute the following formula

$$\varepsilon^{\mu}_{a}\varepsilon^{a,\nu} = ? \tag{18}$$

If we multiply the momentum p_{μ}, it should become vanishing so that the completeness relation should look like

$$\varepsilon^{\mu}_{a}\varepsilon^{a,\nu} = k(-\eta^{\mu\nu} + \frac{p^{\mu}r^{\nu} + p^{\nu}r^{\mu}}{(p\cdot r)^2}) \tag{19}$$

here k is just a overall coefficient.

Here, we choose to compute the completeness relation for polarization vectors with fundamental indices

$$\varepsilon^{\mu}_{IJ}\varepsilon^{IJ,\nu} = a(-\eta^{\mu\nu} + \frac{p^{\mu}r^{\nu} + p^{\nu}r^{\mu}}{(p\cdot r)^2}) \tag{20}$$

with a different coefficient.

For symplisity, we start with the convention from paper arxiv:2202.08257, in which the polarization vector looks like

$$\epsilon^{\mu}_{IJ} = \frac{\langle p_{(I}|\Gamma^{\mu}|r_{J)}\rangle}{\sqrt{2}} \tag{21}$$

with the normalization

$$\langle p^{I}|r^{J}\rangle = \epsilon^{IJ} \tag{22}$$

The method to compute this completeness relation is still just multiplying them togther like

$$\varepsilon^{\mu}_{(IJ)}\varepsilon^{(IJ),\nu} = \frac{\langle p_{(I}|\Gamma^{\mu}|r_{J)}\rangle}{\sqrt{2}} \times \frac{\langle p^{(I}|\Gamma^{\nu}|r^{J)}\rangle}{\sqrt{2}} \tag{23}$$

It can be decomposed into 4 pices IJ^{IJ}, IJ^{JI}, JI^{IJ}, JI^{JI}, and the first one can be computed like

$$\begin{aligned}
\langle p_I|\Gamma^{\mu}|r_J\rangle \times \langle p^I|\Gamma^{\nu}|r^J\rangle &= -\langle p_I|\Gamma^{\mu}|r_J\rangle\langle r^J|\Gamma^{\nu}|p^I\rangle \\
&= -\mathrm{Tr}[|p^I\rangle\langle p_I|\Gamma^{\mu}|r_J\rangle\langle r|^J\Gamma^{\nu}] \\
&= \mathrm{Tr}[(p\cdot\Gamma)\Gamma^{\mu}(r\cdot\Gamma)\Gamma^{\nu}] \\
&= 4(p^{\mu}r^{\nu} + p^{\nu}r^{\mu} - (p\cdot r)\eta^{\mu\nu})
\end{aligned} \tag{24}$$

where we use a nontrival fact that

$$\langle p^I|\Gamma^{\nu}|r^J\rangle = -\langle r^J|\Gamma^{\nu}|p^I\rangle \quad \text{(obtained from} \quad C^T\Gamma^{\mu}C = -(\Gamma^{\mu})^T) \tag{25}$$

The fourth piece is nearly the same as the first one

$$\langle p_J|\Gamma^{\mu}|r_I\rangle \times \langle p^J|\Gamma^{\nu}|r^I\rangle = 4(p^{\mu}r^{\nu} + p^{\nu}r^{\mu} - (p\cdot r)\eta^{\mu\nu}) \tag{26}$$

The remaining two need some extra efforts, here I compute it explictly

$$\langle p_I|\Gamma^{\mu}|r_J\rangle \times \langle p^J|\Gamma^{\nu}|r^I\rangle \tag{27}$$

In terms of two massless spinors that satisfy (22), the $USp(2,2)$ identity operator can be written as

$$|r_a\rangle_A\langle p^a|^B + |p_a\rangle_A\langle r^a|^B = \delta_A^{\ B} \tag{28}$$